Using an MDB with JMS message selectors with WebSphere MQ V7 and WebSphere Application Server V7

IBM Techdoc: 7016581
http://www.ibm.com/support/docview.wss?rs=171&uid=swg27016581

Date last updated: 26-Aug-2009

Angel Rivera – rivera@us.ibm.com
IBM WebSphere MQ Support

+++ Objective +++

To demonstrate the use of a simple Message Driven Bean (MDB) in WebSphere Application Server V7, which interacts with WebSphere MQ V7 as the Java™ Messaging Service (JMS) provider.

This MDB has in its Deployment Descriptor the specification of a JMS message selector for the following user-defined property and value:
color="blue"

This means that this MDB will only receive messages that comply with this selector.

A sample Java program is provided which allows you to create a message with or without a property named "color". You can specify at run time the value for the property. In that way, you can test that the MDB will handle only those messages that comply with the above selector, and those messages that do not comply, are ignored by the MDB.

This document describes the setup and testing of 3 scenarios:

Scenario 1, the MDB "SampleJMSSelectorMdbEjbEAR" is used with a Listener Port. The MDB itself needed to be modified with RAD or another development tool in order to specify the message selector in the Deployment Descriptor. The reason is that the definition of a Listener Port does not have a field for a message selector.

In contrast, in Scenario 2, the same MDB "SampleJMSSelectorMdbEjbEAR" was used with an Activation Specification.
This Activation Specification does NOT handle the message selector specified in the Deployment Descriptor of the MDB. Instead, the message selector is specified in the Activation Specification.

Scenario 3, shows how an MDB which does NOT have a message selector in the Deployment Descriptor and instead, an Activation Specification with a message selector will be used. The MDB is the one described in the referenced techdoc: Using WebSphere MQ V7 as JMS Provider for WebSphere Application Server V7
The name of the MDB is: SampleMDBEJBEAR

+++ Related techdocs and articles +++

1) This techdoc is based on the configuration, deployment and test steps described in the following techdoc.

Using WebSphere MQ V7 as JMS Provider for WebSphere Application Server V7
http://www.ibm.com/support/docview.wss?rs=171&uid=swg27016505

2) The MDB was created with Rational Application Developer (RAD) 7.5 and the Enterprise Archive File (EAR) file which contains the MDB can be downloaded from this techdoc. For more details on how to create and test this MDB, see the following techdoc:

Developing and testing an MDB using RAD 7.5, WebSphere Application Server V7 and MQ V7 as JMS Provider
http://www.ibm.com/support/docview.wss?rs=171&uid=swg27016507

+++ Requisite software +++

The following software was used:

SUSE Linux Enterprise Server (SLES) 9:
  WebSphere Application Server 7.0.0.5
  WebSphere MQ 7.0.0.2
  Firefox (also known as Mozilla)

+++ Downloadable files +++

The following files are included as attachments to this techdoc:

EAR file with MDB:
        SampleJMSSelectorMdbEjbEAR.ear

Text file with code excerpt:
        onMessage-JMSselectors.txt

Java source:
        SampleJMSMsgProperty.java

+ Summary of objects and field values

EJB Project:	SampleJMSSelectorMdbEjb
EAR Project Name:	SampleJMSSelectorMdbEjbEAR
Message-driven bean: "Bean Name":	SampleJMSSelectorMdb
Listener type:	Javax.jms.MessageListener
destinationType:	javax.jms.Queue
WebSphere Bindings:	Listener Port
Listener Port Name:	SampleMDBQueueLP
WebSphere Bindings:	Activation Specification
Act Spec JNDI Name:	jms/SampleMDBQueueActivationSpec
Queue Manager Name:	QM_MDB
Queue Name:	Q_MDB

+++ About the MDB "SampleJMSSelectorMdbEjbEAR"

The MDB was developed as mentioned in the related techdoc:
Developing and testing an MDB using RAD 7.5, WebSphere Application Server V7 and MQ V7 as JMS Provider

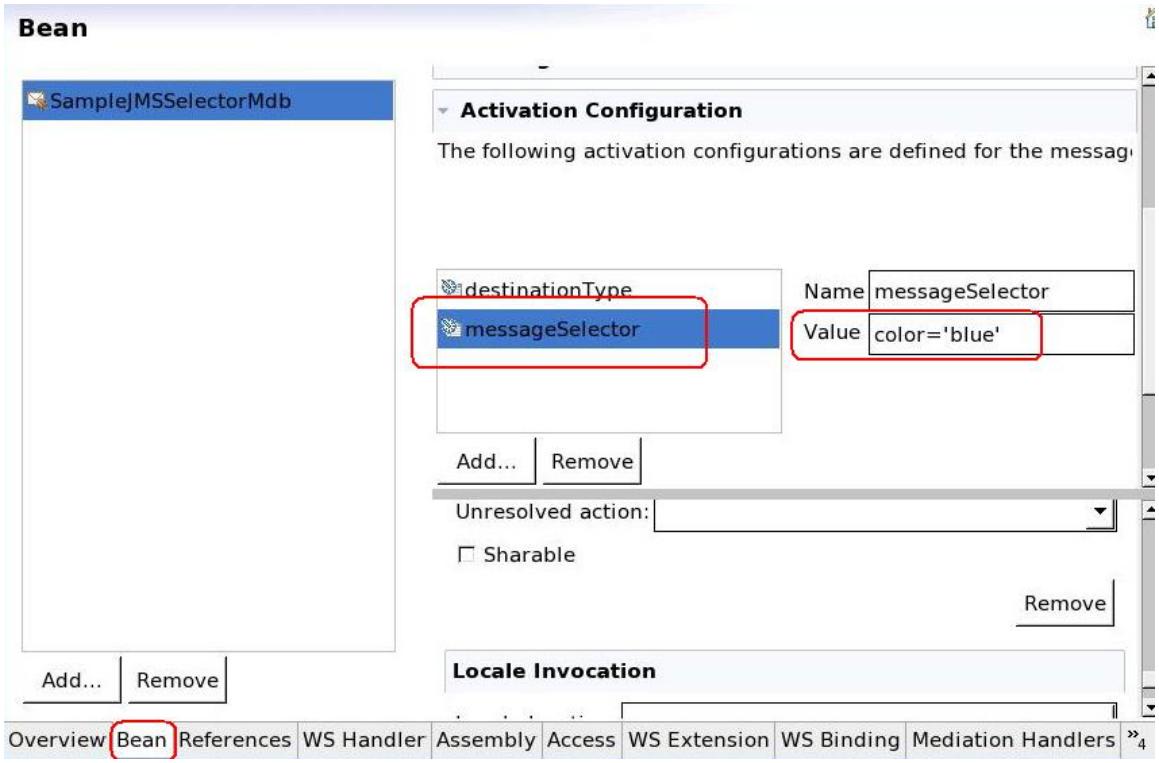
In order to use JMS selectors, it is necessary to modify the Deployment Descriptor for this EJB using RAD. Consult the following web page from the WebSphere Application Server V7 Information Center:

http://publib.boulder.ibm.com/infocenter/wasinfo/v7r0/index.jsp?topic=/com.ibm.websphere.base.doc/info/aes/ae/tmb_depa3.html
Configuring deployment attributes for a message-driven bean against JCA 1.5-compliant resources

The changes are to add a pair of property-value in the SampleJMSSelectorMdb "Bean" tab in the Deployment Descriptor under Activation Configuration:

messageSelector => color='blue'

For this MDB, the specific changes to the Deployment Descriptor look like this:



The onMessage() method of this MDB has the following source code which displays the type of contents (payload) and an “eye catcher string” (+++ SAMPLE MDB) which can let you find quickly the output of the MDB in the SystemOut.log file.

The whole source for this method is available in the following text file associated with this techdoc:

onMessage-JMSselectors.txt

```
public void onMessage(javax.jms.Message msg) {
  try {
    if (msg instanceof javax.jms.TextMessage) {
            System.out.println("+++ SAMPLE JMS-SELECTORS MDB:
Text Message => " + ((javax.jms.TextMessage)msg).getText());
            System.out.println("                       JMS
selector: color => " +
((javax.jms.TextMessage)msg).getStringProperty("color"));
    }
…
```

Notice that the method “getStringProperty” is used to get the value for a property:

```
System.out.println("                       JMS selector: color =>
" + ((javax.jms.TextMessage)msg).getStringProperty("color"));
```

The output in SystemOut.log when a message is processed by this MDB looks like this:

```
SystemOut     O +++ SAMPLE JMS-SELECTORS MDB: Text Message => Text
created by SampleJMSMsgProperty
SystemOut     O                       JMS selector: color => blue
```

++
+++ Setup for Scenario 1: JMS Selectors using Listener Port
++

For further details on the tasks that are mentioned in this section, see the techdoc:
Using WebSphere MQ V7 as JMS Provider for WebSphere Application Server V7

It is assumed in this techdoc that all the objects needed by WebSphere Application Server and by MQ are properly configured.

+ App Server Changes (Part 1)

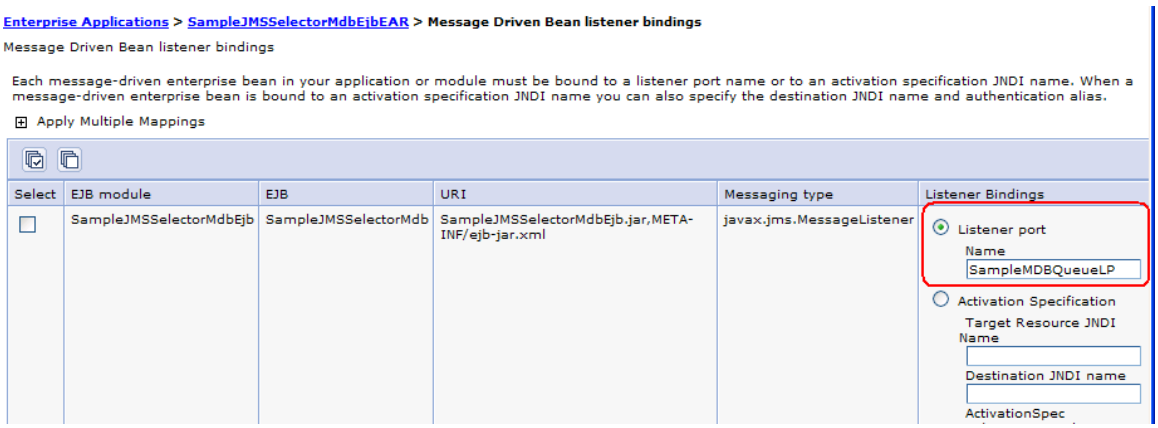
Visit the following page in the Administrative Console:
Applications > New application

Download the sample MDB provided with this techdoc and deploy it.
Use the EAR file: SampleJMSSelectorMdbEjbEAR.ear
At "Step 6: Bind listeners for message-driven beans", specify:
Listener Port: SampleMDBQueueLP

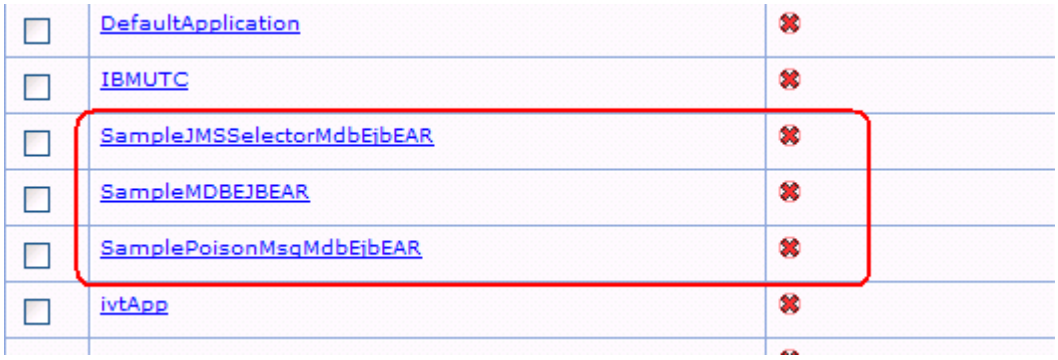
!!! Do NOT start yet the MDB !!!

Verify that the MDB has a listener binding to the Listener Port: SampleMDBQueueLP

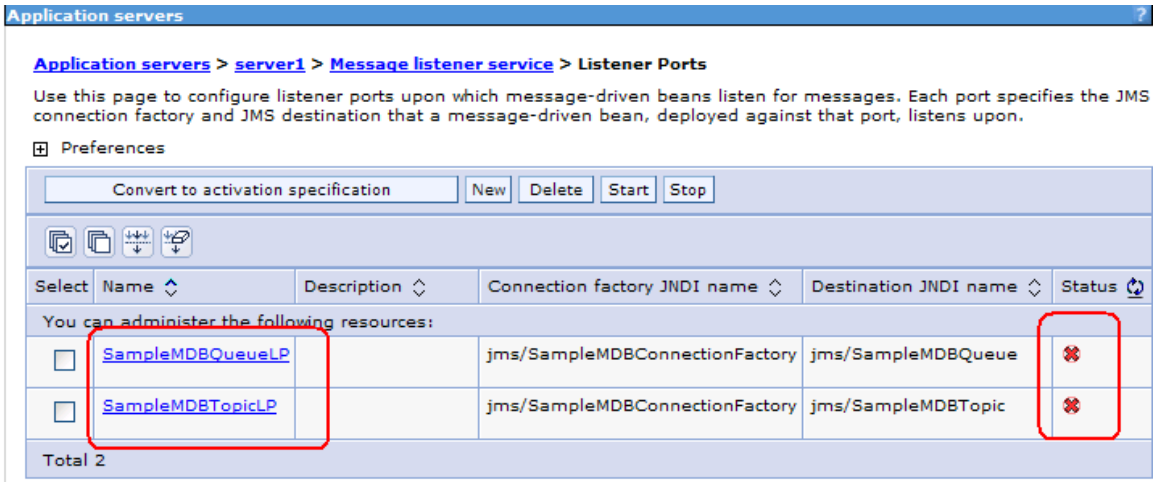
Visit:
Applications > Application Types > WebSphere enterprise applications



To avoid conflicts with related MDBs that consume messages from the same queue, ensure that they are stopped too:



Ensure that the Listener Port is stopped:
SampleMDBQueueLP



+ MQ Changes

Ensure that the queue “Q_MDB” from the queue manager “QM_MDB” is empty. You can accomplish it by issuing the following MQ sample that will read all the messages from the queue:

```
$ amqsget Q_MDB QM_MDB
```

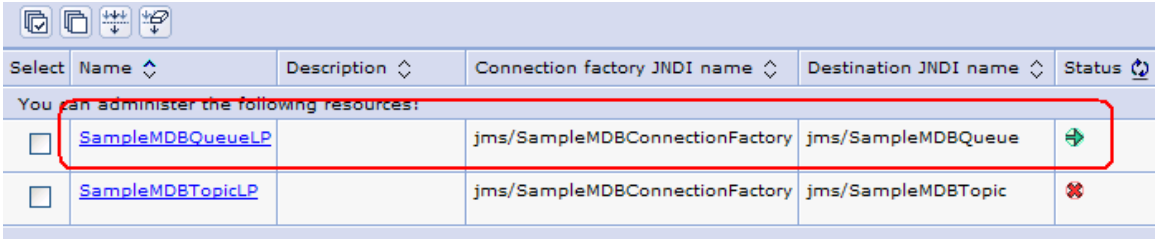
Note: This sample terminates after 30 seconds have passed without receiving new messages. It is OK to terminate it with Ctrl-C.

Issue the following command to verify that the queue is empty. Notice that CURDEPTH has a value of 0.

```
$ echo "display ql(Q_MDB)" | runmqsc QM_MDB | grep CURDEPTH
   CURDEPTH(0)                                  DEFBIND(OPEN)
```

+ App Server Changes (part 2)

Start the Listener Port: SampleMDBQueueLP



Start the EJB: SampleJMSSelectorMdbEjbEAR

+ Update the Java sample source code

In order to keep the Java sample source code as simple as possible, the connection information is hard-coded. Thus, you will need to modify the source code to suit your needs and compile it.

It explicitly sets the hostname to: localhost
… and the server-connection channel is: SYSTEM.DEF.SVRCONN

Run the following MQ script to setup the proper Java and JMS variables for MQ.
NOTE: Ensure to invoke it with a dot then a blank space!

```
$ . setjmsenv
MQ_JAVA_INSTALL_PATH is /opt/mqm/java
MQ_JAVA_DATA_PATH is /var/mqm
MQ_JAVA_LIB_PATH is /opt/mqm/java/lib
CLASSPATH
is :/opt/mqm/java/lib/com.ibm.mq.jar:/opt/mqm/java/lib/com.ibm.mq
jms.jar:/opt/mqm/samp/java/base:/opt/mqm/samp/java/jms
```

Modify the file “SampleJMSMsgProperty.java” as described in the comments in the header of the file. There are 3 lines that you need to update.
Note that in this example the port 1420 was used to avoid a conflict with an existing queue manager that already uses the default 1414 port number. You can use specify another port number for the MQ Listener for your queue manager.

```
…
    cf.setIntProperty(WMQConstants.WMQ_PORT, 1420);
…
    cf.setStringProperty(WMQConstants.WMQ_QUEUE_MANAGER,
"QM_ANGELILLO");
…
    destination = session.createQueue("queue:///Q1");
```

Additional lines that you may want to review:

```
    cf.setStringProperty(WMQConstants.WMQ_HOST_NAME, "localhost");
…
    cf.setStringProperty(WMQConstants.WMQ_CHANNEL,
"SYSTEM.DEF.SVRCONN");
```

Compile the program:

```
$ javac SampleJMSMsgProperty.java
```

You should not get any warnings or errors.

++
+++ Testing the Scenario 1: JMS Selectors using Listener Port
++

Let's test the MDB. We will issue 3 commands to generate 3 messages. Only one of them should be handled by the MDB (the one with color='blue')

A: No color
$ java SampleJMSMsgProperty
* Creating message with no color

B: color='red'
$ java SampleJMSMsgProperty red
* Creating message with color red

C: color='blue'
$ java SampleJMSMsgProperty blue
* Creating message with color blue

SystemOut.log should show:
SystemOut O +++ SAMPLE JMS-SELECTORS MDB: Text Message => Text created by SampleJMSMsgProperty
SystemOut O JMS selector: color => blue

We will need 2 UNIX command prompt windows:

+ Window 1: One for watching the recent entries in the WebSphere Application Server SystemOut.log. We can watch for the output from the MDB that indicates that a message was received.

+ Window 2: The other for entering an MQ command that places one message into the queue that is monitored by the WebSphere Application Server Listener Port and which passes the message to the MDB.

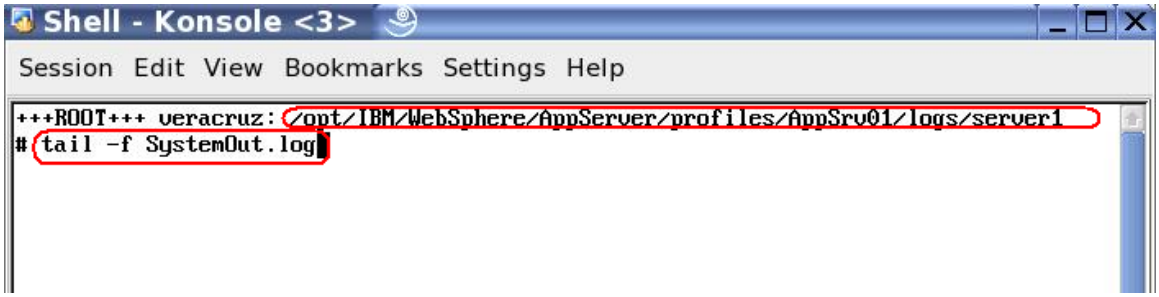
From Window 1:

Change to the directory where the WebSphere Application Server server logs reside. In this case it is:

```
$ cd /opt/IBM/WebSphere/AppServer/profiles/AppSrv01/logs/server1
```

Issue the command to watch constantly the recent lines into the SystemOut.log file.

```
$ tail -f SystemOut.log
```



A) Case: message with NO property

Expected result: The MDB should NOT receive this message because it does not comply with the message selector: color='blue'

From Window 2:

Login as user "mqm" (or another user who has access to MQ).

Execute the recently compiled program.
Do NOT specify any parameters. This means that the color property will not be set.

```
$ java SampleJMSMsgProperty
* Creating message with no color
Sent message:

  JMSMessage class: jms_text
  JMSType:          null
  JMSDeliveryMode:  2
  JMSExpiration:    0
  JMSPriority:      4
  JMSMessageID:
ID:414d5120514d5f4d44422020202020200d7c7c4a02400520
  JMSTimestamp:     1251144207643
  JMSCorrelationID: null
  JMSDestination:   queue:///Q_MDB
  JMSReplyTo:       null
  JMSRedelivered:   false
    JMSXAppID: WebSphere MQ Client for Java
    JMSXDeliveryCount: 0
    JMSXUserID: rivera
    JMS_IBM_PutApplType: 28
    JMS_IBM_PutDate: 20090824
    JMS_IBM_PutTime: 20032767
Text created by SampleJMSMsgProperty
SUCCESS
```

In Window 1 notice that there are NO messages displayed by the MDB.

Remember that the message selector is
  color='blue'
… and the above message did not have the property "color" defined at all.

B) Case: message with property: color='red'

Expected result: The MDB should NOT receive this message because it does not comply with the message selector: color='blue'

From Window 2:

Specify the color 'red':

```
$ java SampleJMSMsgProperty red
* Creating message with color red
Sent message:

  JMSMessage class: jms_text
  JMSType:          null
  JMSDeliveryMode:  2
  JMSExpiration:    0
  JMSPriority:      4
  JMSMessageID:
ID:414d5120514d5f4d44422020202020200d7c7c4a02420520
  JMSTimestamp:     1251144513664
  JMSCorrelationID: null
  JMSDestination:   queue:///Q_MDB
  JMSReplyTo:       null
  JMSRedelivered:   false
    JMSXAppID: WebSphere MQ Client for Java
    JMSXDeliveryCount: 0
    JMSXUserID: rivera
    JMS_IBM_PutApplType: 28
    JMS_IBM_PutDate: 20090824
    JMS_IBM_PutTime: 20083368
    color: red
Text created by SampleJMSMsgProperty
SUCCESS
```

Notice that the last property shown above is:
    color: red

In Window 1, you can verify that the MDB has not displayed any messages.

C) Case: message with property: color='blue'

Expected result: The MDB should receive this message because it does comply with the message selector: color='blue'

From Window 2:

Specify the color 'blue':

```
$ java SampleJMSMsgProperty blue
* Creating message with color blue
Sent message:

  JMSMessage class: jms_text
  JMSType:          null
  JMSDeliveryMode:  2
  JMSExpiration:    0
  JMSPriority:      4
  JMSMessageID:
ID:414d5120514d5f4d44422020202020200d7c7c4a02440520
  JMSTimestamp:     1251144635701
  JMSCorrelationID: null
  JMSDestination:   queue:///Q_MDB
  JMSReplyTo:       null
  JMSRedelivered:   false
    JMSXAppID: WebSphere MQ Client for Java
    JMSXDeliveryCount: 0
    JMSXUserID: rivera
    JMS_IBM_PutApplType: 28
    JMS_IBM_PutDate: 20090824
    JMS_IBM_PutTime: 20103572
    color: blue
Text created by SampleJMSMsgProperty
SUCCESS
```

Notice that the last property shown above is:

```
    color: blue
```

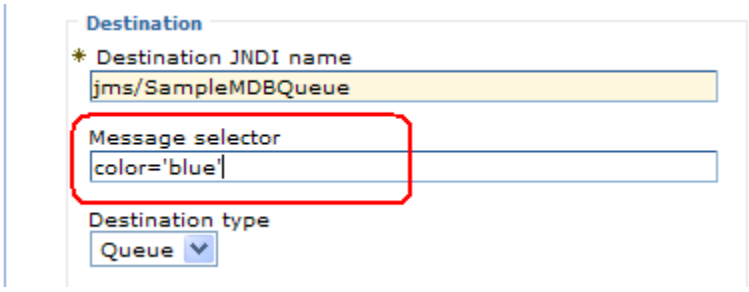
In Window 1, you can verify that the MDB has not displayed any messages.

```
SystemOut     O +++ SAMPLE JMS-SELECTORS MDB: Text Message
=> Text created by SampleJMSMsgProperty
SystemOut     O                        JMS selector: color
=> blue
```

++
+++ Setup for Scenario 2: JMS Selectors using Activation Specification
++

When using an Activation Specification, the message selector can be entered in the resource definition from the Administrative Console.

Visit:
Resources > JMS > Activation specifications
Select SampleMDBQueueActivationSpec
Scroll down to the section "Destination"
Enter the desired string for "Message selector"
  color='blue'



Then click on OK and click on Save.

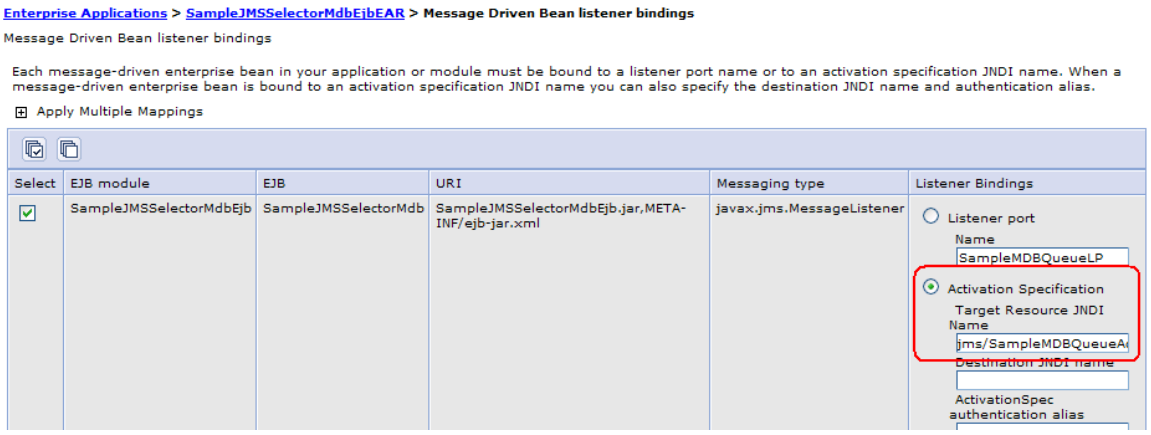
DO NOT RESTART THE SERVER YET!!! Let's do few more changes first.

Stop the application in order to change the listener binding from a Listener Port to an Activation Specification.

Visit:
Applications > Application Types > WebSphere enterprise applications
Select: SampleJMSSelectorMdbEjbEAR
Then Stop.

Change the listener binding for the MDB to be the following Activation Specification:
        jms/SampleMDBQueueActivationSpec

Select again the application.
Visit:
Enterprise Applications > SampleJMSSelectorMdbEjbEAR > Message Driven Bean listener bindings



Then click on OK and click on Save.

Now you can stop the server and restart it.

Just to prove that the Listener Port is not involved, let's stop it.
Visit:
Servers > Server Types > WebSphere application servers
Select server1
Application servers > server1 > Message listener service > Listener Ports
Select SampleMDBQueueLP
Then click on Stop.

Name	Description	Connection factory JNDI name	Destination JNDI name	Status
can administer the following resources:				
SampleMDBQueueLP		jms/SampleMDBConnectionFactory	jms/SampleMDBQueue	

To avoid conflict with other MDBs that handle messages from this test queue, ensure that only the following MDB is running and stop the others.

Visit:
Applications > Application Types > WebSphere enterprise applications
Select the MDBs to be stopped. Click on Stop.
Only the following MDB should be running:
SampleJMSSelectorMdbEjbEAR

SampleJMSSelectorMdbEjbEAR	
SampleMDBEJBEAR	
SamplePoisonMsgMdbEjbEAR	

++
+++ Testing the Scenario 2: JMS Selectors using Activation Specification
++

Repeat the steps from “Testing Scenario 1”.

The expected results are exactly the same as in Scenario 1.

++
+++ Setup for Scenario 3: Using Activation Specification with another MDB
++

Notice that in Scenario 1, an MDB was used with a Listener Port. The MDB itself needed to be modified with RAD or another development tool in order to specify the message selector in the Deployment Descriptor. The reason is that the definition of a Listener Port does not have a field for a message selector.

In contrast, in Scenario 2, the MDB was used with an Activation Specification. This Activation Specification does NOT handle the message selector specified in the Deployment Descriptor of the MDB. Instead, the message selector is specified in the Activation Specification.

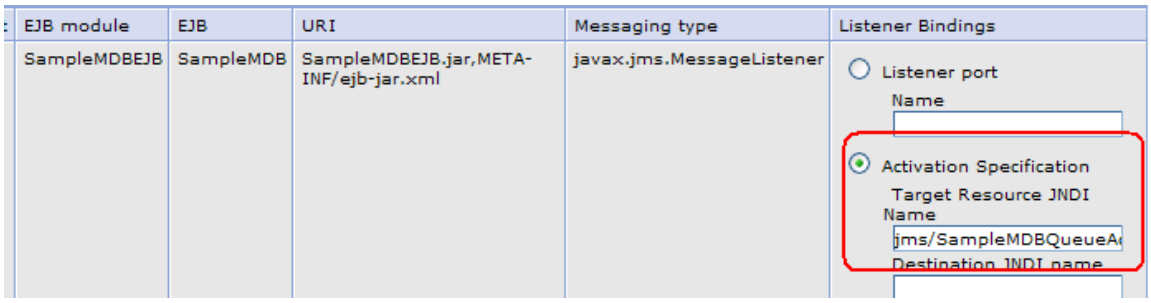
The purpose for this Scenario 3 is to use an MDB which does NOT have a message selector in the Deployment Descriptor and instead, an Activation Specification with a message selector will be used. The MDB is the one described in the referenced techdoc:
Using WebSphere MQ V7 as JMS Provider for WebSphere Application Server V7
The name of the MDB is:
    SampleMDBEJBEAR

This is one of the advantages of using Activation Specifications over a Listener Port: a message selector can be specified via the Administrative Console and there is no need to modify the Deployment Descriptor of the MDB via RAD.

Let's stop the application: SampleJMSSelectorMdbEjbEAR
Applications > Application Types > WebSphere enterprise applications
Select: SampleJMSSelectorMdbEjbEAR
Then Stop

Notice that we will be using another MDB!!

Change the listener binding for the application:
    SampleMDBEJBEAR
Use the following Act Spec:
    jms/SampleMDBQueueActivationSpec



EJB module	EJB	URI	Messaging type	Listener Bindings
SampleMDBEJB	SampleMDB	SampleMDBEJB.jar,META-INF/ejb-jar.xml	javax.jms.MessageListener	Listener port Name Activation Specification Target Resource JNDI Name jms/SampleMDBQueueA Destination JNDI name

Then click OK and Save.

Restart the application:
Applications > Application Types > WebSphere enterprise applications
Select: SampleMDBEJBEAR
Then Start

Proceed to test it.

++
+++ Testing the Scenario 3: Using Activation Specification with another MDB
++

Repeat the steps from “Testing the Scenario 1”.

The expected results are exactly the same as in Scenario 1.
HOWEVER, this MDB “SampleMDBEJBEAR” does not have the code to display the property “color”, thus, the MDB will only display this line:

```
SystemOut     O +++ SAMPLE MDB: Text Message => Text
created by SampleJMSMsgProperty
```

This is the end of the techdoc.

+++ end +++